Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	25 October 2026
Team ID	EN25CS3010534
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Python (Flask), HTML, CSS, Tableau calculated fields	Tableau calculated fields (a Year extraction field and a KPI Selector field) do the calculations for medals, KPIs and efficiency. Python with Flask runs the web application, and HTML and CSS build the pages that show the dashboards.
2	Database / Data Warehouse	Warehouse used to store and manage structured business data for analysis and reporting.	CSV files: athletes.csv (8,000 rows), medal_results.csv (910 rows), games_summary.csv (18 rows), alltime_country_rankings.csv (60 rows)	The data is small, structured and kept in flat files, so no database server is needed. CSV files are easy to share, store on GitHub and reload into Tableau.
3	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau Data Connections (Text File)	Tableau connects directly to the four CSV files with no extra loading step. The files are related on noc_code and year so that all data sits in one data source.
4	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau Desktop (Public Edition): relationships, calculated fields, KPI Selector parameter	Used to check data types, nulls and duplicates, relate the four files, extract the Year field, and calculate KPIs such as athlete participation, nations represented and average medal efficiency. The KPI Selector parameter switches a chart between Total, Gold, Silver and Bronze medals.
5	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau: KPI cards, world map, line, bar, pie, bubble and donut charts, 2 dashboards and a story	Tableau is a standard BI tool. Its dashboards and story make a large Olympic dataset easy to explore and explain.
6	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public, Flask website (HTML/CSS template)	Tableau Public hosts the published dashboards and story with their filters and parameters. The Flask website embeds them in pages for Home, Insights, Dashboard 1, Dashboard 2 and Story, so anyone can use them in a browser.